Bell Work Answer Key

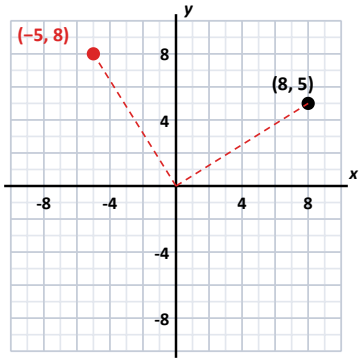
Wednesday, October 28 · Module 2, Topic 2, Lesson 3, Day 1 · Slopes of Perpendicular Lines

1. What is the reciprocal of $^{2}/_{3}$?

A. −2/3 **B. 3/2 ✓** **C.** −3/2 **D.** 1

Flip the fraction: 2/3 · 3/2 = 1.

2. Rotate the point (8, 5) 90° counterclockwise about the origin. Where does it go?



A. (5, −8) **B.** (−8, −5) **C. (−5, 8) ✓** **D.** (8, −5)

A 90° counterclockwise turn sends (x, y) to (−y, x): (8, 5) → (−5, 8). SE p. 306.

3. What is the slope of the segment from (0, 0) to (8, 5)?

A. 8/5 **B.** −5/8 **C.** 13 **D. 5/8 ✓**

Slope = rise ÷ run = (5 − 0) ÷ (8 − 0) = 5/8. SE p. 306.

4. A line passes through the points (−1, 4) and (2, −2). What is the slope of the line?

F. 2 **G. −2 ✓** **H.** −1/2 **J.** 1/2

Slope = (−2 − 4) ÷ (2 − (−1)) = −6 ÷ 3 = −2.

Go over: Q1 (reciprocal) sets up today's rule: perpendicular slopes are **flipped and the sign changed**. Q2–Q3 preview the Lesson 3 video: the segment to (8, 5) has slope 5/8; after the turn, (0, 0) to (−5, 8) has slope −8/5.